

REVIEW OF DIFFERENTIATION

Rules

1. **Constant:** $\frac{d}{dx} c = 0$

2. **Constant Multiple:** $\frac{d}{dx} cf(x) = c f'(x)$

3. **Sum:** $\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$

4. **Product:** $\frac{d}{dx} f(x)g(x) = f(x)g'(x) + g(x)f'(x)$

5. **Quotient:** $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

6. **Chain:** $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$

7. **Power:** $\frac{d}{dx} x^n = nx^{n-1}$

8. **Power:** $\frac{d}{dx} [g(x)]^n = n[g(x)]^{n-1} g'(x)$

Functions

Trigonometric:

9. $\frac{d}{dx} \sin x = \cos x$

10. $\frac{d}{dx} \cos x = -\sin x$

11. $\frac{d}{dx} \tan x = \sec^2 x$

12. $\frac{d}{dx} \cot x = -\csc^2 x$

13. $\frac{d}{dx} \sec x = \sec x \tan x$

14. $\frac{d}{dx} \csc x = -\csc x \cot x$

Inverse trigonometric:

15. $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$

16. $\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$

17. $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$

18. $\frac{d}{dx} \cot^{-1} x = -\frac{1}{1+x^2}$

19. $\frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$

20. $\frac{d}{dx} \csc^{-1} x = -\frac{1}{|x|\sqrt{x^2-1}}$

Hyperbolic:

21. $\frac{d}{dx} \sinh x = \cosh x$

22. $\frac{d}{dx} \cosh x = \sinh x$

23. $\frac{d}{dx} \tanh x = \operatorname{sech}^2 x$

24. $\frac{d}{dx} \coth x = -\operatorname{csch}^2 x$

25. $\frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$

26. $\frac{d}{dx} \operatorname{csch} x = -\operatorname{csch} x \coth x$

Inverse hyperbolic:

27. $\frac{d}{dx} \sinh^{-1} x = \frac{1}{\sqrt{x^2+1}}$

28. $\frac{d}{dx} \cosh^{-1} x = \frac{1}{\sqrt{x^2-1}}$

29. $\frac{d}{dx} \tanh^{-1} x = \frac{1}{1-x^2}$

30. $\frac{d}{dx} \coth^{-1} x = \frac{1}{1-x^2}$

31. $\frac{d}{dx} \operatorname{sech}^{-1} x = -\frac{1}{x\sqrt{1-x^2}}$

32. $\frac{d}{dx} \operatorname{csch}^{-1} x = -\frac{1}{|x|\sqrt{x^2+1}}$

Exponential:

33. $\frac{d}{dx} e^x = e^x$

34. $\frac{d}{dx} a^x = a^x (\ln a)$

Logarithmic:

35. $\frac{d}{dx} \ln|x| = \frac{1}{x}$

36. $\frac{d}{dx} \log_a x = \frac{1}{x(\ln a)}$